

KARA RIPPE

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PROFESSIONAL PROFILE

Undergraduate Psychology and Data Science student with a developing focus on Human Factors and Human-Computer Interaction. Combines R/Python with experimental psychology to design interfaces that reduce cognitive workload and increase engagement in digital and spatial environments.

EDUCATION

University of Kansas | Lawrence, KS

Bachelor of Arts in Psychology | GPA: 3.96 | Expected May 2027

- Minor: Classical Antiquity | Certificate: Interdisciplinary Data Science
- Honors: Dean's List (2023–present), Academic Honors Scholarship
- Methodological Coursework: Research Methods in Psychology, Foundations of Data Science (R), Data 1: Dealing with Data (Python), Moral Issues in Computer Technology (Technology Ethics)

RESEARCH & INTERACTION DESIGN PROJECTS

Analysis of Aerospace Mission Dynamics & Media Tropes | *Quantitative Researcher & Data Analyst* | *Spring 2026*

- Collaborated with a 4-person research team to evaluate global aerospace launch success data across time (1957-2022)
- Wrangled, cleaned, and transformed raw data on international aerospace launches (N = 4,324) in R to model machine reliability and historical failure rates
- Produced data visualizations using ggplot2, including custom tables, bar graphs, and time-series trendlines

Player Engagement and Media Tropes | *Quantitative Researcher & Data Analyst* | *Spring 2026*

- Analyzed *Backlogg'd* user data within a 4-person research team to compare player logging behaviors and genre trends across roughly forty years of game releases (1980-2022)
- Situated findings within broader media trope patterns to examine company success across genres, allowing useful insight into what components players found most meaningful

- Translated these findings into tables with targeted color coding (*gt*), comparable trendlines (*patchwork*), and a dumbbell plot

Cognitive Load Optimization for Spatial Video-Game Displays | *Interaction Designer* | *Independent Project, Spring 2026*

- Constructed a low-fidelity Head-Up Display (HUD) conceptual design adapted for time-compressed, 3D simulated environments
- Repositioned vital resource indicators (health/mana bars) to the upper visual periphery with the goal of diminishing cognitive workload during high-stress tasks
- Designed a centralized lock-on mechanic paired with context-sensitive UI elements that automatically decay during passive exploration to prevent visual masking

Nexus Mods Front-Page Usability Redesign | *UX Researcher & Interface Designer* | *Independent Project, Spring 2026*

- Collected qualitative user data about the Nexus Mods' website from community forums to establish a baseline for a redesign in Figma, focusing on visual architecture attuned to returning users
- Restructured the primary landing interface from dense vertical arrays into easily scannable horizontal grids to prioritize immediate access to frequently used and personalized content, including favored games, easy uploading, and user-curated mods
- Applied visual ergonomics principles by adjusting typography weight and luminance contrast across the signature orange/grey color palette to prevent visual fatigue

TECHNICAL & METHODOLOGICAL SKILLS

- Languages & Software: R (RStudio, tidyverse, ggplot2, dplyr), Python (Pandas, NumPy, Google Colab), Figma, Excel, Foundational HTML/CSS
- Research Methodology & HCI: Cognitive Workload Optimization, Usability Evaluation, Information Architecture, Hypothesis Testing, Inferential Statistics, Data Cleaning